

Design Opportunities

What the Week 1 research scan found, and how it should change the system.

System Conversational Memory Intelligence System

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Format 20-minute design review (not a paper presentation)

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Companions component_scan.md,
idea_evaluation_matrix.md, challenge_notes.md

Backlog ../design_backlog.md · ADR-001/002/003

THE ONE SLIDE

The scan confirmed the Deliverable 1 design more than it overturned it, with one sharp addition. **Adopt a bi-temporal validity model and move conflict resolution off the ranker** (external evidence: Zep). **Adopt an evaluation scaffold** so the non-negotiable baseline comparison is buildable (LongMemEval). **Adopt a testable PII admission gate** (Presidio). Prototype one admission mechanism (Mem0). Defer or reject the rest, including one SOTA result that breaks a core invariant. And a real 2026 paper argues the whole system is unnecessary — it is answered, and it improved the plan.

1. Where the design was weakest entering the week

Deliverable 1 is a first-principles specification with **no measured numbers**. The scan targeted its four soft spots rather than the whole landscape (handbook §4.2 step 1):

GAP	WHAT WAS UNDER-SPECIFIED OR UNPROVEN	CAPABILITY
G1	Admission mechanism: <i>how</i> store-vs-update-vs-discard and PII gating are actually done	C2
G2	Temporal validity asserted from first principles, with no external evidence and no measurement	C5, C3
G3	Consolidation — the highest-risk component (F5/F6) — had no chosen mechanism	C7
G4	No evaluation set exists, yet baseline comparison is a non-negotiable gate	C10

2. Decisions at a glance

IDEA	SOURCE	FIXES	DECISION	ONE-LINE REASON
Bi-temporal validity + invalidation	Zep 2501.13956	F4 (G2)	ADOPT	Fixes the failure that justifies C5; low runtime cost
LongMemEval eval scaffold	LongMemEval 2410.10813	G4	ADOPT	De-risks the whole measurement chain; imports abstention
Presidio PII gate	Presidio (OSS)	F11 (G1)	ADOPT	Turns a policy into a tested, deterministic control
Mem0 operation selector	Mem0 2504.19413	F8 (G1)	PROTOTYPE	Concrete update-vs-add, but LLM-per-candidate cost is unproven <i>here</i>
A-MEM memory evolution	A-MEM 2502.12110	—	DEFER/REJECT	Mutation breaks C7 cite-don't-replace → reintroduces F5/F6
LLMLingua compression	LLMLingua 2310.05736	F2	REJECT V1	Marginal at a 2k-token budget over compact typed facts

3. Highest-value opportunity: bi-temporal validity (adopt)

PROBLEM ADDRESSED

F4 — a superseded fact served as current. **INFERENCE** `first_principles.md §6` makes the entire case for multi-signal ranking (C5, the component that "does the most work") hinge on this failure being real.

MECHANISM

Each typed fact carries *valid-from* / *valid-to* (world time) and *recorded-at* / *invalidated-at* (system time). A contradicting fact **invalidates** the prior one (sets its valid-to) instead of deleting it. Retrieval filters to currently-valid facts; history is kept for audit and re-derivation.

EVIDENCE, AND HOW FAR IT TRANSFERS

VERIFIED Zep reports up to +18.5% on LongMemEval and DMR 94.8% vs 93.4% against MemGPT, crediting temporal handling. **INFERENCE** The setting (assistant chat + business data) is closer to GTM than Generative Agents' simulation but still not multi-tenant or regulated, so the *mechanism* transfers while the *magnitude* does not.

THE REAL DESIGN MOVE

Not just a schema field: **relocate conflict resolution from the ranker to the write path**. The ranker then only ever orders facts that are all currently valid, which is cleaner than asking a relevance+recency score to encode validity — something `failure_analysis.md` D-1 argued it structurally cannot do.

WHY ADOPT THE MODEL BUT NOT THE ENGINE

The full Graphiti temporal knowledge graph would violate the v1 non-goal of hand-specified, inspectable structure and add heavy operational surface. The bi-temporal *fields* and *invalidation semantics* capture the benefit at a fraction of the cost.

VALIDATION (BOUNDED, IN DELIVERABLE 3)

Three-arm comparison on the fixed set: naive similarity vs similarity+recency vs validity-filter. Metric: how often a superseded preference is served over the current one.

WHAT WOULD REVERSE IT

TO MEASURE If similarity+recency matches the validity-filter on the fixed set, P4 does not force this in practice and it is downgraded to defer. This is falsifier 1 from Deliverable 1, now wired into the Day-3 protocol.

4. Two supporting adoptions

4.1 LongMemEval as the evaluation scaffold (adopt taxonomy + subset, extend)

C10 needs a fixed adversarial set and baseline comparison is a non-negotiable gate. LongMemEval's five abilities include **knowledge updates** (= F4) and **abstention** (= cold-start / no-relevant). **INFERENCE** Adopt its taxonomy and case designs as the backbone; it has **no** cross-tenant (F10) or PII (F11) abilities, so those GTM-critical cases are authored on top. Guard: never imply its coverage extends to isolation. (ADR-002.)

4.2 Presidio PII gate at admission (adopt deterministic-first)

F11 is unrecoverable and needs a *tested* control. **INFERENCE** Deterministic recognizers first (cheap, no LLM on the PII path, no injection surface), NER optional and measured. Recognizer recall is imperfect, so deletion (C7) stays as the backstop — the gate is defence-in-depth, not a sole guarantee. (ADR-003.)

5. The challenge that made the plan better

STEELMAN — "NONE OF THIS IS NECESSARY"

VERIFIED *Back to Basics* (arXiv:2604.11628, 2026) argues the bottleneck is a *Signal Sparsity Effect*, not memory architecture, and that a minimalist retrieval+generation pipeline beats stronger memory baselines on token/latency cost. **INFERENCE** It does **not** refute this design: its benchmarks are recall tasks; it says nothing about F10 cross-tenant leakage, F11 incidental PII, or F9 tested deletion — the unrecoverable failures that motivate the system. A better retriever cannot satisfy an isolation invariant at any accuracy.

But it changed two things. (1) The Day-3 baseline must be a *strong* minimalist retriever (Turn-Isolation Retrieval), not a straw man — a fairer, harder baseline to beat. (2) It sharpened falsifier 1: if the minimalist retriever already handles knowledge-updates on the fixed set, the temporal model is not earning its cost and gets cut. I now hold the temporal model as *adopt-pending-measurement*, not near-certain.

6. What lands where

INTO DELIVERABLE 3 (BASELINE)

- 3-arm comparison (naive / +recency / validity)
- Strong baseline retriever (Turn-Isolation)
- Prototype + cost-measure the Mem0 selector
- Fixed set on the LongMemEval scaffold + GTM cases

INTO DELIVERABLE 4 (DESIGN)

- Bi-temporal schema + invalidation (ADR-001)
- Ranker orders only currently-valid facts (B-005)
- Evaluation taxonomy (ADR-002)
- Admission PII gate + deletion backstop (ADR-003)

Sources

1. Rasmussen, P., et al. *Zep: A Temporal Knowledge Graph Architecture for Agent Memory*. 2025. arXiv:2501.13956
2. *Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory*. 2025. arXiv:2504.19413
3. Xu, W., et al. *A-MEM: Agentic Memory for LLM Agents*. 2025. arXiv:2502.12110
4. Wu, D., et al. *LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory*. ICLR 2025. arXiv:2410.10813
5. Wu, Y., et al. *Back to Basics: Let Conversational Agents Remember with Just Retrieval and Generation*. 2026. arXiv:2604.11628
6. Jiang, H., et al. *LLMLingua: Compressing Prompts for Accelerated Inference of LLMs*. 2023. arXiv:2310.05736

7. Microsoft Presidio. <https://github.com/microsoft/presidio>

AI assistance disclosure. Claude helped locate candidate sources and draft this review. The relevance judgments, the adopt/prototype/defer/reject decisions, the position changes recorded in `challenge_notes.md`, and every verified / inference / assumption / to-measure boundary are mine and are what I will defend in the design review.